SMART PREGNANCY PAIN-RELIEF BELT USING SENSOR-BASED VIBRATION AND HEAT THERAPY

MD. Zakaria Islam Mahi
ID: 2011009

Upama Devi
ID: 2011011

Rudra Chakraborty
ID: 2011026

Submitted to: Ayesha Binth Humayun
Department of Biomedical Engineering

I. Introduction

Pregnancy-related lower back and pelvic pain is a highly prevalent musculoskeletal condition, affecting approximately 70% of pregnant women, particularly during the second and third trimesters. This pain arises due to a combination of biomechanical loading, postural changes, and hormonal effects, which lead to ligament laxity and muscle strain.

Specifically, the hormone relaxin causes ligament laxity, while the anterior shift of the center of gravity increases lumbar lordosis by approximately 30–40°. In addition, fetal weight loading (up to 4 kg) significantly alters spinal biomechanics. These changes result in abnormal spinal loading, where L5–S1 disc pressure increases by 25–30%, activating nociceptors in paraspinal muscles and ligaments and triggering inflammation through prostaglandin pathways.

Persistent pain negatively affects mobility, sleep quality, work performance, and mental well-being, especially among working women. In Bangladesh, pregnant women in labor-intensive sectors such as textile and garment industries experience high pain levels (typically 6–8 out of 10), leading to reduced productivity and an increased risk of preterm labor (odds ratio 1.5–2.0).

Although conventional maternity belts provide passive mechanical support, they do not address the neurophysiological mechanisms of pain. Furthermore, pharmacological pain management during pregnancy is limited due to potential fetal risks, highlighting the need for safe alternatives.

Currently, there is a lack of affordable, wearable, home-based, and non-invasive pain management devices. Therefore, there is a critical need for a drug-free, safe, and effective solution that can actively modulate pain signals and provide real-time relief without clinical supervision.

To address these limitations, this work proposes a smart pregnancy belt that delivers sensor-triggered vibration and heat therapy to relieve lower back pain through neurophysiological and circulatory mechanisms.

The system is based on the Gate Control Theory of Pain, where non-painful sensory stimuli (such as vibration or warmth) activate fast-conducting nerve fibers, which inhibit the transmission of slower pain signals at the spinal cord level. The system utilizes a DHT22 sensor for temperature detection and a force-sensitive resistor (FSR) for measuring pressure distribution. These sensors are directly interfaced with an ESP32 microcontroller, which performs real-time data acquisition and processing without the need for an external Arduino board, thereby reducing system complexity and power consumption.

The proposed system provides an affordable, scalable, and real-time solution for monitoring physiological conditions, with potential applications in maternal healthcare, ergonomics, rehabilitation, and workplace wellness. By integrating biomechanical sensing with wireless communication in a unified system framework, it enables early detection and management of discomfort, contributing to improved maternal well-being and reduced health risks.

II. Used Components

The proposed Body Sensor Network (BSN) system is implemented using low-cost, readily available hardware components that enable real-time acquisition of biomechanical and physiological signals. The primary components used in this project are described as follows:

A. ESP32 Microcontroller

The ESP32 serves as the central processing unit of the system. It is a low-power, high-performance microcontroller with integrated Wi-Fi and Bluetooth capabilities. The ESP32 is responsible for acquiring analog signals from all connected sensors, performing data processing, and executing the decision-making algorithm. In this implementation, the ESP32 is used directly without an external Arduino board, reducing system complexity and improving efficiency.

B. DHT22 Temperature Sensor

The DHT22 is a digital temperature and humidity sensor used to monitor physiological temperature conditions in real time. It provides calibrated digital output, ensuring high accuracy and stability compared to analog sensors. The sensor measures temperature using an internal thermistor and transmits the data through a single-wire communication protocol

C. Force Sensitive Resistor (FSR)

The force sensitive resistor (FSR) is used to measure pressure distribution on the seating surface. The resistance of the FSR decreases with increasing applied force. In this system, the FSR is configured as part of a voltage divider circuit to produce an analog voltage proportional to the applied pressure. Uneven pressure distribution is interpreted as a potential indicator of poor sitting posture.

D. Vibration Motor

The vibration motor serves as the primary actuator in the system.Two vibration motors can be independently controlled to provide localized or distributed stimulation depending on the detected pressure intensity. Since the ESP32 cannot supply sufficient current to drive the motors directly, each motor is interfaced through a transistor-based driver circuit. An NPN transistor (e.g., 2N2222) is used as a switch, where the ESP32 GPIO controls the transistor base through a resistor, allowing the motor to be powered from an external supply. A flyback diode is connected across the motor terminals to protect the circuit from back electromotive force (EMF) generated during switching.

E. Resistors and Supporting Components

A fixed resistor (typically $10\,\mathrm{k}\Omega$) is used in conjunction with the force-sensitive resistor (FSR) to form a voltage divider circuit, enabling stable and accurate analog signal acquisition by the ESP32. Additionally, a pull-up resistor ($10\,\mathrm{k}\Omega$) is commonly used with the DHT22 temperature sensor to ensure reliable digital communication. A $220\,\Omega$ resistor is also employed in series with the LED to limit current and protect the component from damage.Other supporting components include jumper wires, breadboard connections, and regulated power supply lines, which are essential for proper interfacing and stable operation of the sensors and actuators with the ESP32 microcontroller.

F. Power Supply

The system is powered using the ESP32's 3.3,V supply, ensuring compatibility with all connected analog sensors. Proper grounding is maintained across all components to ensure stable and noise-free signal acquisition.

III. Methodology

The proposed system introduces a smart pregnancy pain-relief belt that integrates sensor-based monitoring, intelligent control, and therapeutic actuation to alleviate lower back and pelvic pain during pregnancy. The system is designed based on the Gate Control Theory of Pain, where non-noxious stimuli such as vibration and heat are utilized to modulate pain perception through neurophysiological and circulatory mechanisms.

The architecture employs an ESP32 microcontroller for real-time sensor acquisition, local processing, and wireless communication. The system combines pressure sensing, temperature monitoring, automated therapy activation, and IoT-based control into a unified wearable solution.

A. Hardware Architecture and Interfacing

All sensing and actuation components are interfaced with the ESP32 microcontroller. A DHT22 sensor is connected to a digital input pin to monitor skin temperature, primarily for safety and feedback control. A force-sensitive resistor (FSR) is connected to an analog input pin through a voltage divider circuit to measure pressure variations caused by body movement or localized muscle tension.

The therapeutic components include vibration motors and a flexible heating element. These are connected to digital output pins through MOSFET or transistor-based driver circuits to ensure safe current handling, as the ESP32 cannot directly drive high-current loads. The ESP32 continuously reads sensor data, processes control logic, and actuates the corresponding therapy modules.

B. Sensor Data Acquisition and Processing

The system continuously acquires physiological and mechanical signals:

- **Temperature (°C):** Measured using the DHT22 sensor for thermal monitoring and safety regulation.
- **Pressure (Analog FSR value):** Represents localized force variations associated with posture changes or muscle tension.

Sensor data are periodically sampled and processed using threshold-based logic. Instead of relying on fixed physiological constants, the system uses experimentally calibrated thresholds to detect relative changes such as increased mechanical strain or abnormal pressure variations. These inputs act as triggers for adaptive therapeutic responses.

C. Vibration Therapy Module

The vibration therapy module is activated based on pressure variations detected by the FSR sensor. Different intensity levels are defined to provide graded stimulation:

- Low pressure ($<$1500): No stimulation
- Moderate pressure (1500–3000): Motor 1 activated
- High pressure ($>$3000): Motor 2 activated

The vibration motors operate within a frequency range of approximately 100–200 Hz to stimulate cutaneous mechanoreceptors.

Neurophysiological Mechanism:

According to the Gate Control Theory of Pain, stimulation of large-diameter A-beta mechanoreceptive fibers inhibits nociceptive signals transmitted by smaller A-delta and C fibers at the spinal cord level (substantia gelatinosa). This reduces the transmission of pain signals to the brain, resulting in a natural analgesic effect without pharmacological intervention.

D. Heat Therapy Module

The heat therapy module provides controlled thermal stimulation to the lower back region to promote muscle relaxation and improve blood circulation.

A temperature sensor is used in a closed-loop configuration to ensure safe operation. The ESP32 continuously compares the measured temperature with predefined thresholds.

The lamp is controlled automatically based on the measured temperature. The ESP32 compares the real-time temperature value with a predefined threshold. If the measured temperature exceeds 29°C, the lamp is turned ON. Otherwise, the lamp remains OFF. To improve reliability, the system first checks whether the temperature reading is valid. If the sensor fails to return a valid measurement, the lamp is forced into the OFF state. This prevents incorrect switching due to invalid or undefined sensor data. The heating system is designed to operate within a therapeutic temperature range (approximately 38–40°C) with automatic cutoff to ensure user safety.

Physiological Effects:

- Promotes vasodilation and increased blood circulation
- Reduces muscle stiffness and spasms
- Enhances tissue oxygenation
- Provides relief from musculoskeletal discomfort

E. Control Logic and Decision System

The embedded control algorithm operates in real time and includes:

- Continuous sensor data acquisition
- Threshold-based condition evaluation
- Automatic activation of vibration and heat therapy
- Safety monitoring (temperature limits and sensor validation)
- Wireless data transmission and optional remote control via IoT interface

The system dynamically adapts therapy delivery based on real-time physiological feedback, ensuring safe, personalized, and effective pain management.

F. Design Considerations and Scientific Validity

The proposed system is based on established biomedical principles, including neuromodulation, thermal therapy, and wearable assistive design. Unlike conventional maternity belts that provide passive support, this system actively modulates pain pathways using controlled stimulation.

It is important to note that there is no universally defined external threshold for fetal kick force or pain intensity measurable through surface sensors. Therefore, the system relies on relative pressure variations rather than absolute force values, and calibration is performed experimentally.

Similarly, temperature sensing is primarily used for safety regulation rather than direct pain detection. The integration of these subsystems provides a novel, non-pharmacological, and wearable solution for managing pregnancy-related discomfort.

IV. Circuit Diagram

Figure 4 shows the complete circuit diagram of the proposed Body Sensor Network (BSN) system. The diagram illustrates how the ESP32 microcontroller is interfaced with the DHT22 sensor, force-sensitive resistor (FSR), vibration motor, and LED.

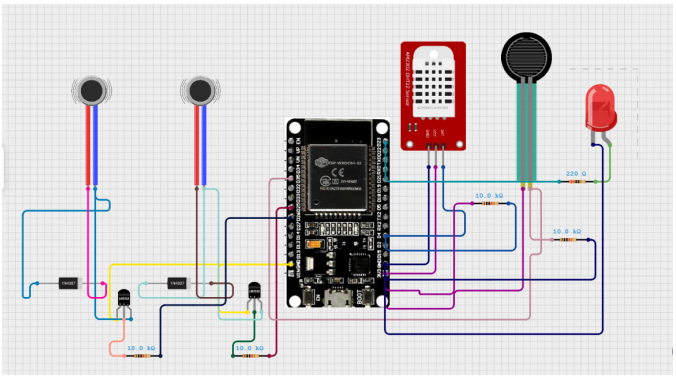


Fig. 1: Circuit diagram of the proposed smart pregnancy belt system

V. Remote Monitoring Integration

The proposed smart pregnancy belt system incorporates remote monitoring capabilities using Wi-Fi-enabled communication through the ESP32 microcontroller. The ESP32 acts as a lightweight web server that continuously collects real-time physiological and pressure data from the integrated sensors, including the DHT22 temperature sensor and the force-sensitive resistor (FSR).

The acquired data is processed locally and transmitted over a wireless network in the form of structured text data via HTTP requests. A mobile application developed using MIT App Inventor retrieves this data by sending periodic GET requests to the ESP32 server using its IP address.

The received data string, containing parameters such as temperature, pressure (FSR), lamp status, and motor states, is parsed within the mobile application using string-splitting techniques. These values are then displayed on the user interface in real time, allowing continuous monitoring of the user's condition.

In addition to monitoring, the system supports remote control functionality. The mobile application sends control commands (e.g., `/motor1/on`, `/motor2/off`, `/auto`) to the ESP32 server, enabling users to manually activate vibration therapy or switch between automatic and manual modes.

VI. Experimental Setup Image and Prototype Implementation

The prototype of the proposed system was developed using an ESP32 microcontroller integrated with a DHT22 temperature sensor, force-sensitive resistor (FSR), vibration motors, and a heating element.

The experimental setup was assembled on a breadboard for initial testing, where all sensors and actuators were properly interfaced with the ESP32. The system was powered using a regulated power supply to ensure stable operation. Sensor readings were continuously monitored, and corresponding outputs were observed in real time.

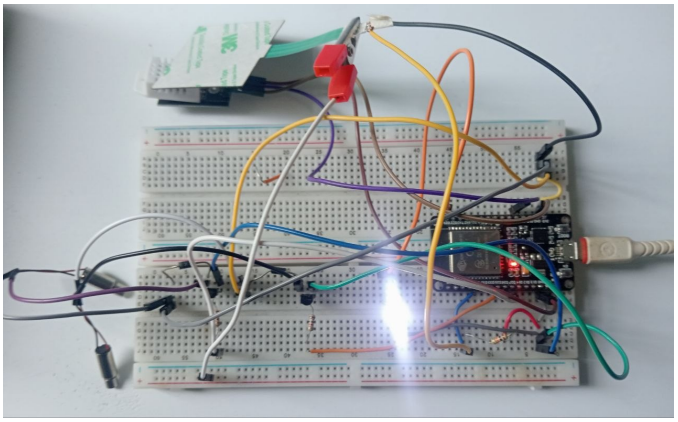


Fig. 2: Real-time implementation of smart pregnancy belt system



Fig. 3: MIT App Inventor implementation of the smart pregnancy belt application

VII. Cost Analysis

The proposed system is designed to be cost-effective and suitable for widespread use, particularly in low-resource settings. The total estimated cost of the prototype is relatively low due to the use of readily available and affordable components.The overall system cost remains within an affordable range, making it accessible for a wide population, especially in developing countries. The use of low-cost microcontrollers and sensors significantly reduces production expenses without compromising system functionality.

Among all components, the force-sensitive resistor (FSR) contributes the highest cost, indicating a potential area for future cost optimization through alternative sensing technologies or bulk procurement. The remaining components, including the ESP32 and supporting hardware, are economically viable and widely available in the local market.

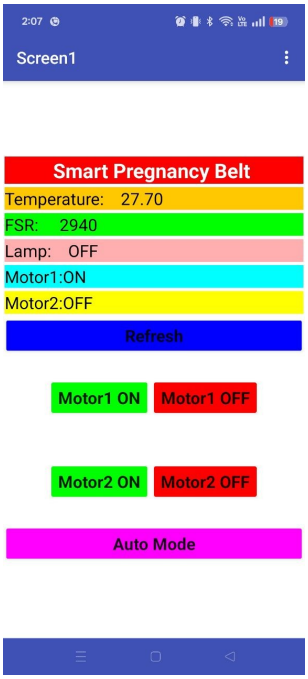


Fig. 4: MIT App Inventor implementation of the smart pregnancy belt application

Furthermore, the cost can be further minimized through large-scale production, custom PCB design, and integration of components into a compact wearable module. This would reduce wiring complexity and assembly cost.

TABLE I: Cost Analysis of Proposed System

Component	Approximate Cost (BDT)
ESP32 Microcontroller	450
DHT22 Sensor	350
Force Sensitive Resistor (FSR)	1800
Vibration Motors (2 units)	200
Heating Element	300
Supporting Components (resistors, wires, etc.)	200
Power Supply	200
Total Estimated Cost	**3500 BDT**

VIII. Future Work

The proposed smart pregnancy pain-relief belt demonstrates a promising approach to non-invasive and real-time pain management. However, several improvements can be explored in future work to enhance system performance and usability.

Future development may include the integration of additional physiological sensors such as electromyography (EMG) for muscle activity monitoring and heart rate sensors for comprehensive maternal health assessment.also to invent a low-cost, self-made sensor fr better use which can cover a large area. Machine learning algorithms can be incorporated to

analyze sensor data patterns and provide personalized therapy recommendations based on user-specific conditions.

Further miniaturization of hardware components and flexible wearable design can improve comfort and long-term usability. The inclusion of rechargeable battery systems and energy-efficient power management techniques will enhance portability.

Additionally, cloud-based data storage and advanced IoT integration can enable long-term monitoring, data analytics, and remote medical supervision. Clinical validation through trials involving pregnant users is also essential to evaluate the effectiveness, safety, and reliability of the system in real-world conditions.

IX. Conclusion

This paper presents the design and implementation of a smart pregnancy pain-relief belt that integrates sensor-based monitoring with automated vibration and heat therapy. The system is developed based on established biomedical principles such as the Gate Control Theory of Pain and thermal therapy mechanisms.

By utilizing real-time data from pressure and temperature sensors, the system provides adaptive and non-pharmacological pain relief, ensuring safety and comfort for pregnant users. The integration of IoT-based remote monitoring further enhances usability by enabling real-time tracking and control through a mobile application.

The proposed solution addresses the limitations of conventional maternity belts by offering an active, intelligent, and affordable alternative for managing pregnancy-related discomfort. Overall, the system demonstrates strong potential for improving maternal health, especially in resource-constrained environments.

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